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Infrared and Raman spectra of AA-stacking bilayer graphene

Yuehua Xu, Xiaowei Li and Jinming Dong[1]

Group of Computational Condensed Matter Physics, National Laboratory of Solid State Microstructures and Department of Physics, Nanjing University, Nanjing 210093, People's Republic of China

E-mail: jdong@nju.edu.cn

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Abstract

The infrared (IR) absorption and resonant Raman spectra of the undoped AA-stacking bilayer graphene have been calculated using density-functional theory in the local density approximation. It is found that the undoped AA-stacking bilayer graphene exhibits a different characteristic jump and peak structure in its IR spectra, and a different G-band peak structure in its resonant Raman spectra, compared with those of the AB-stacking one, which are caused by the different interlayer coupling effects, due to different stacking types in both of them. Based upon the different IR and Raman spectra, a powerful experimental method can be proposed to identify accurately the stacking type and the layer number in future experiments.

1. Introduction

Graphene, a flat monolayer of carbon atoms, packed tightly into a two-dimensional (2D) honeycomb lattice, is a basic building block for all other graphitic materials, which was believed to be unstable in the free state [1] and so described as an 'academic' material previously. However, monolayer graphene (MLG) was unexpectedly made several years ago [2] and other few-layer graphenes (few-LGs) were also obtained recently. The graphene exhibits exceptionally high crystal and electronic quality, leading to a major research effort into its growth [3], structure [4] and phonon [4–6] and electronic excitations [5], optical spectra [6, 7] and transport properties [8], which have already revealed a lot of new physics and potential applications.

A graphite crystal is a vertical stack of hexagonal graphene sheets along their *c* axis, coupled by the van der Walls interaction. There are three different stacking modes of the graphene layers, including simple hexagonal, orthorhombic and rhombohedral systems, i.e. AA-, AB-and ABC-stacking graphite. In general, natural graphite adopts the AB-stacking sequence because a small energy barrier exists between the AB-and AA-stacking ones [9]. However, there are also some theoretical and experimental studies on the AA-stacking sequence and rhombohedral graphite with the ABC-stacking sequence [10–13].

[1] Author to whom any correspondence should be addressed.

For the AB-stacking bilayer graphene (BLG), it is found that the interlayer coupling destroys the symmetry and isotropy of its energy bands, making the MLG linear bands become parabolic ones, which remarkably causes a weak overlap between its valence and conduction bands [14–18]. In the past few years, there are many theoretical and experimental works on the electronic and optical properties of the AB-stacking few-LGs, especially their infrared (IR) and Raman spectra. For example, their IR spectra show that their low-energy dispersions markedly change with increasing layer number due to the interlayer couplings [19–24]. Also the experimental Raman spectra of the AB-stacking few-LGs [25, 26] exhibit unique signatures for the layer number. In particular, the experimental first-order Raman spectra of the AB-stacking BLG mainly concern the high-energy Raman G-band [27, 28]. Meanwhile, there are several related theoretical works on their Raman spectra [29–31].

On the other hand, there is little theoretical work on the AA-stacking BLG [32, 33] due to a lack of practical samples and related experimental observations. Recently, however, Lee *et al* [34] reported evidence of growing the AA-stacking graphite on (111) diamond. Very recently, Liu *et al* [35] surprisingly found that the BLG often exhibits AA stacking, which is very hard to distinguish from the MLG. Consequently, it is necessary to do more research, both theoretically and experimentally, on the electronic excitations and optical properties of the AA-stacking BLG because they should strongly depend on the interlayer couplings and so the stacking sequences.

Therefore, in this paper, we will perform first-principles calculations on the IR absorption spectra and the first-order resonant Raman spectra for the undoped AA-stacking BLG, and pay more attention to their differences with those of the AB-stacking BLG.

2. Theory and computational detail

2.1. Theory of IR absorption and resonant Raman scattering

By diagonalizing the Hamiltonian matrix, the energy dispersion $E_h(\vec{k})$ and the related eigenfunction $|\psi_h(\vec{k})\rangle$ of the system can be obtained, where $h = \mathrm{c}(h = \mathrm{v})$ represents the unoccupied (occupied) state. Furthermore, the IR absorption coefficient of a few-LG system can be obtained by the following formula:

$$A(\omega) \propto \sum_{\mathrm{c,v},j_1,j_2} \int_{\mathrm{1stBZ}} \frac{\mathrm{d}^2\vec{k}}{(2\pi)^2} \times \left| \langle \psi_\mathrm{c}(\vec{k}, j_1) \left| \frac{\vec{E}\cdot\vec{P}}{m_\mathrm{e}} \right| \psi_\mathrm{v}(\vec{k}, j_2)\rangle \right|^2 \times \{ f[E_\mathrm{c}(\vec{k}, j_1)] - f[E_\mathrm{v}(\vec{k}, j_2)]\} \times \delta[E_\mathrm{c}(\vec{k}, j_1) - E_\mathrm{v}(\vec{k}, j_2) - \omega]. \tag{1}$$

The j_1, j_2 are indices of the conduction and valence subbands, respectively, counted from the Fermi energy E_F. $\vec{E}$ is the direction of the photon's electric field and ω is its frequency. $\vec{P}$ is the electron's momentum operator and m_e is its mass. $f[E_{\mathrm{c(v)}}(\vec{k}, j_{1(2)})]$ is the Fermi–Dirac distribution function. Under an electric field, parallel to the graphene layers, i.e. $\vec{E} \parallel x (\vec{E} \parallel y)$, or perpendicular to the graphene layers, i.e. parallel to the stacking direction $\vec{E} \parallel z$, the electrons could be excited from the occupied valence π bands to the unoccupied conduction π^* bands. At zero temperature ($T = 0$ K), only the inter-π-band excitations can occur. The excitation energy is $\omega_\mathrm{ex} = E_\mathrm{c}(\vec{k}) - E_\mathrm{v}(\vec{k}) = \omega$, due to the almost zero of the photon's momentum, and hence the optical selection rule is $\Delta\vec{k} = 0$.

The resonant Raman intensity for the first-order G-band scattering in the Stokes process is calculated following the standard formula given in [36] and using the VASP code [37].

2.2. Computational details

In our calculations, we have used the total-energy plane-wave pseudopotential method in the local density approximation (LDA) with the Ceperley–Alder exchange correlation. The Vienna *ab initio* simulation package (VASP) [37] is employed. The interaction between the ions and electrons is described by the highly accurate full-potential projected augmented wave (PAW) [38, 39], which can give a more accurate and reliable result than the ultrasoft potential. In our calculations, the 2s and 2p orbitals of the carbon atom are treated as the valence orbitals and a large plane-wave cutoff of 500 eV is used throughout.

A supercell geometry is adopted, so that the BLGs are aligned in a supercell, with the closest distance between the adjacent BLGs being at least 10 Å along the stacking direction, which is taken along the z direction. A uniform grid of $n \times n \times 1$ k-points is used, and the maximum spacing of k-points is less than 0.01 Å^{-1}.

We first construct an ideal AA- or AB-stacking BLG from the graphite. Then, a conjugated gradient method is used to optimize the atomic positions, and also the supercell's length and width, until the force acting on all the atoms is less than 0.005 eV Å^{-1}. With the equilibrium structure, we then used a more dense set of k-points, i.e. 2596 k-points in our energy zone of concern, to calculate the electronic structure and the transition matrix elements, and finally we used equation (1) to obtain the IR absorption coefficient.

In order to calculate the resonant Raman intensity, besides the electronic structure and the optical transition matrix, we also need to calculate the phonon's normal modes at the Γ-point of the BLGs.

3. Results and discussions

The geometrical structures of the AA- and AB-stacking BLGs are shown in figures 1(a) and (b), respectively, for which there exist two atoms of A_i and B_i in a primitive cell of each graphene. The C–C bond length a is 1.41 Å in both of the AA- and AB-stacking BLGs, which is almost the same as the corresponding C–C bond length in the bulk graphite. But the distance b between the two layers is somewhat different in both of them, which is 3.59 Å for the AA-stacking BLG and 3.31 Å for the AB-stacking one. Our LDA calculation results for the b values are close to the experimental ones (3.55 Å for the AA-stacking graphite [34] and 3.35 Å for the AB-stacking one [40]).

Their low-energy dispersions are given, respectively, in figures 1(c) and (d). The low-energy bands are linear and almost isotropic near the K-point, whose slopes are very close to that of the MLG. The interlayer coupling only makes the BLGs' energy bands intersect each other and its Fermi energy changed from the K-point to the neighboring two points. For the AB-stacking BLG, the interlayer coupling obviously changes the MLG linear bands as the parabolic ones near the K-point, causing further an overlap between its valence and conduction bands, which destroys the symmetry between its valence and conduction bands about E_F.

The calculated IR absorption spectra of the undoped AA- and AB-stacking BLGs are shown in figures 2(a) and (b), respectively, for (a) $\vec{E} \parallel x$ and (b) $\vec{E} \parallel z$, where only those optical transitions from the valence bands to the conduction ones are allowed. In addition, their corresponding joint densities of states (JDOS) are given in figure 2(c). It is clearly seen from figure 2 that: (1) their IR spectra show a strong anisotropy for (a) $\vec{E} \parallel x$ and (b) $\vec{E} \parallel z$, which is particularly obvious for the AA-stacking BLG. For example, there exists one jump at about 450 meV in the IR spectrum of the AA-stacking BLG when $\vec{E} \parallel x$, as shown in figure 2(a), which is mainly caused by the transitions between those nonparallel sections of the valence band and conduction band, denoted by the blue arrows in figure 2(d). On the other hand, when

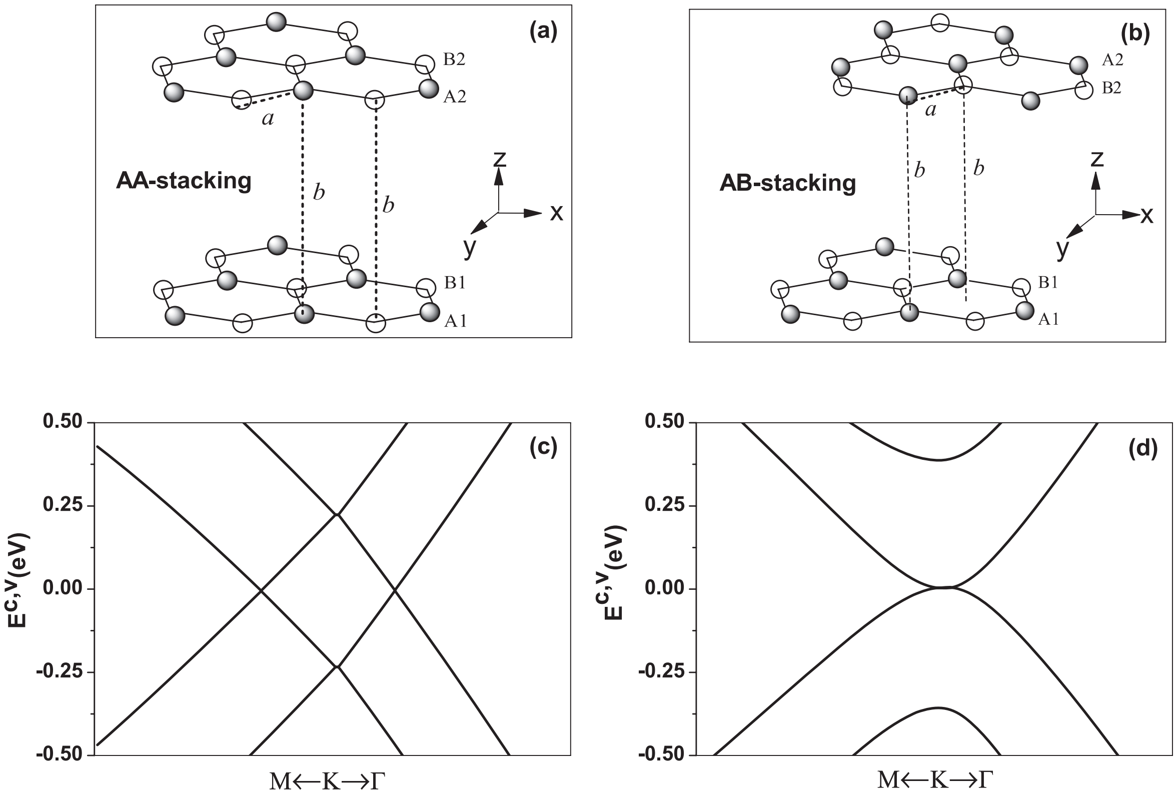

Figure 1. (a) and (b) Schematic of the AA- and AB-stacking bilayer graphenes. Here, a is the C–C bond length in the graphene and b is the distance between two layers. Their low-energy dispersions are shown, respectively, in (c) for the AA stacking and (d) for the AB stacking.

$\vec{E} \parallel z$, a peak at 450 meV appears in its IR spectrum, as shown in figure 2(b), which is induced by the corresponding peak in its JDOS, shown in figure 2(c). That is because there are other transition channels between those parallel sections of the valence band and conduction band, denoted by the dashed black arrows in figure 2(d). (2) For the AB-stacking BLG, there also exists one jump at about $\omega = 373$ meV when $\vec{E} \parallel x$, which is also mainly caused by the transitions between those nonparallel band sections of the valence band $j_2 = -2(-1)$ and conduction band $j_1 = 1(2)$, denoted by the blue arrows in figure 2(e). That is also because its interlayer coupling lifts the energy degeneracy around the K-point, inducing an average energy level splitting, which is about 373 meV, as shown in figures 1(d) and 2(e). However, when $\vec{E} \parallel z$, a jump, instead of the peak, appears at about $\omega = 746$ meV, which is caused by those transitions between its valence band $j_2 = -2$ and the conduction one $j_1 = 2$ near the K-point with their energy difference being about 746 meV, denoted by the dashed black arrows in figure 2(e).

Obviously, the IR absorption spectra of the AA- and AB-stacking BLGs are very different from each other. For example, in the case of $\vec{E} \parallel x$, although both of them exhibit a characteristic big jump, their positions lie at different frequencies, which are 450 and 373 meV, respectively for the AA-stacking BLG and the AB-stacking one. Moreover, the difference between their IR spectra is more obvious when $\vec{E} \parallel z$. For example, a characteristic peak appears at 450 meV in the IR spectrum of the AA-stacking BLG, but in contrast, a jump lies at 746 meV in that of the AB-stacking one. Thus, a powerful experimental tool could be proposed to identify accurately the AA- and AB-stacking sequence by their different IR spectra.

On the other hand, although it is very hard to distinguish the AA-stacking BLG from a single graphene layer in the experiment [35], our calculations indicate that the MLG only shows a basically featureless IR spectrum, which is caused by its linear energy bands at the K-point. So, the significant difference between the IR spectra of the AA-stacking BLG and MLG could provide a useful experimental method to identify the AA-stacking BLG from the MLG, which is similar to the case of the AB-stacking one.

The calculated first-order Stocks resonant G-band Raman spectra of the undoped AA- and AB-stacking BLGs are shown, respectively, in figures 3(a) and (b). It is clearly seen from figure 3 that: (1) the AA-stacking BLG exhibits two pairs of resonant peaks, lying at 3.95, 4.15 eV (E_{-21}, $E_{-21} + E_G$) and 4.11, 4.31 eV (E_{-12}, $E_{-12} + E_G$), respectively. Here, the $E_{-21}(E_{-12})$ is the optical transition energy from the valence band $j_2 = -2(-1)$ to the conduction one $j_1 = 1(2)$, as shown in figure 4. $E_G = \hbar\omega_G$ is the G-band phonon energy, being 0.1973 eV (1591.2 cm^{-1}). (2) But the AB-stacking BLG shows only one pair of resonant peaks, lying at 4.09 and 4.29 eV ($E_{-11}(E_{-22})$, $E_{-11}(E_{-22}) + E_G$). Here, $E_G = \hbar\omega_G$ is 0.1974 eV (1592.4 cm^{-1}), which is bigger by about 1.2 cm^{-1} than that of the AA-stacking one. This conclusion is well consistent with other theoretical calculation results [31]. Obviously, the first-order resonant Raman spectra

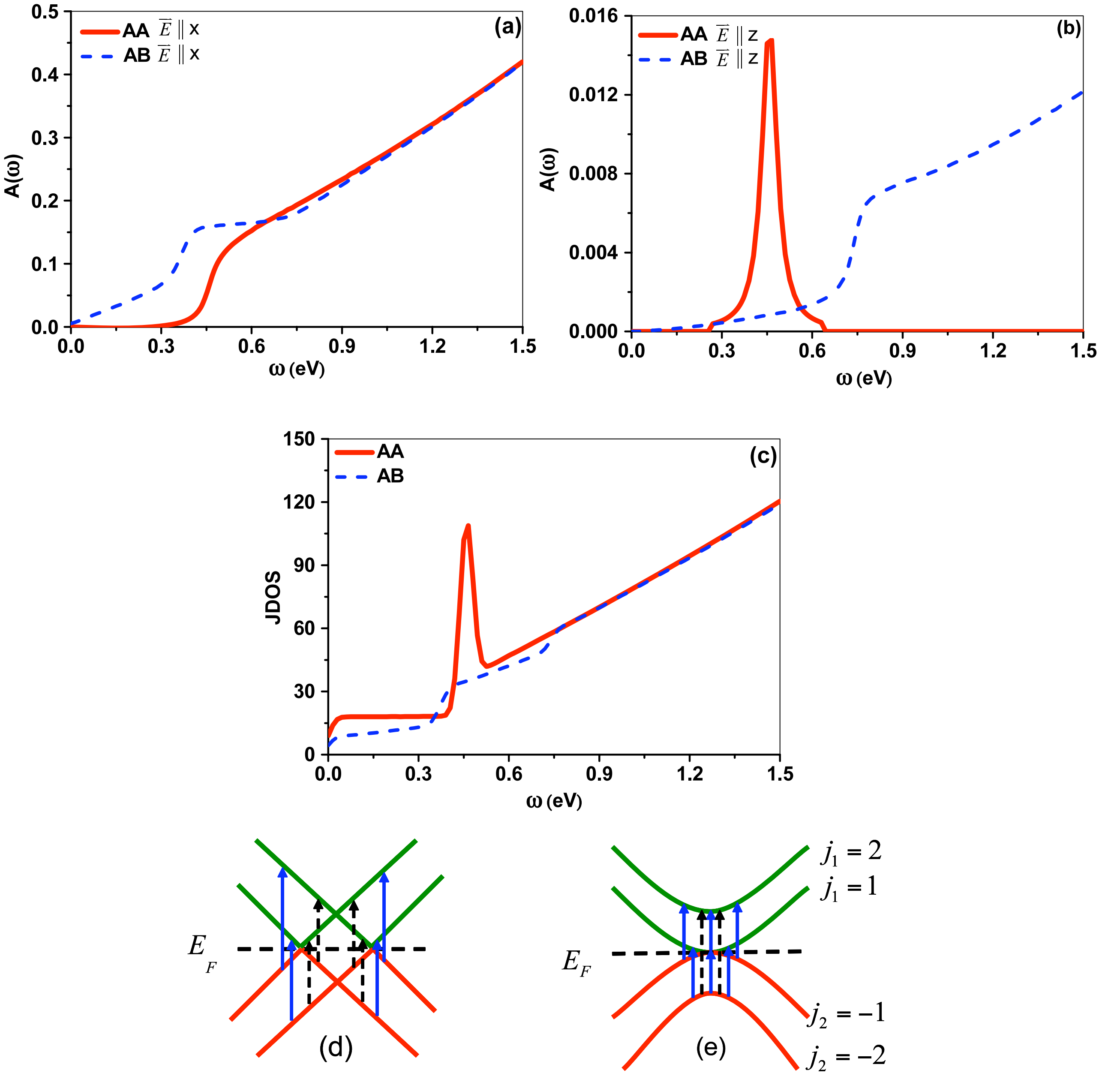

Figure 2. The calculated IR absorption spectra of the AA- and AB-stacking BLGs when (a) $\vec{E} \parallel x$ and (b) $\vec{E} \parallel z$, and (c) their corresponding joint densities of states (JDOS). Schematic shows of their band structures near the K-point are given in (d) for the AA stacking, and (e) for the AB stacking. Here, both blue and dashed black arrows denote the permitted optical transitions.

(This figure is in colour only in the electronic version)

of both the AA- and AB-stacking BLGs are very different from each other, which could also be used to identify both of them experimentally.

In order to understand why there exists so large a difference between the first-order Raman spectra of the AA- and AB-stacking BLGs, we have shown their energy band structures along $M \rightarrow K$ in figures 4(a) and (b), respectively. It is seen clearly from figure 4 that: (1) for the AA-stacking BLG, from its valence band $j_2 = -2$ to its conduction band $j_1 = 1$, its optical transition energy E_{-21} is smaller by 0.16 eV than the E_{-12} (from its valence band $j_2 = -1$ to its conduction band $j_1 = 2$), making so one optical absorption peak to split into two, lying at E_{-21} and E_{-12}, respectively. Thus, their corresponding G-band Raman peaks lie at $E_{-21} + E_G$ and $E_{-12} + E_G$, respectively. (2) But, for the AB-stacking BLG, its two conduction bands $j_1 = 1$ and 2 cross somewhere, so making its two permitted optical excitations to come from $j_2 = -1$ to $j_1 = 1$ and from $j_2 = -2$ to $j_1 = 2$, which have almost the same energy, i.e. $E_{-11} = E_{-22}$. Therefore, only one pair of doubly degenerate first-order resonant G-band Raman peaks could be found at $E_{-11}(E_{-22})$, $E_{-11}(E_{-22})+E_G$, respectively. As for their resonant peak intensity, it can be seen from figures 4(a) and (b) that the intensity of the AA-stacking BLG is nearly half of that for the AB-stacking one, which is caused by the breaking of the double degeneracy for the optical transitions of the AA-stacking BLG.

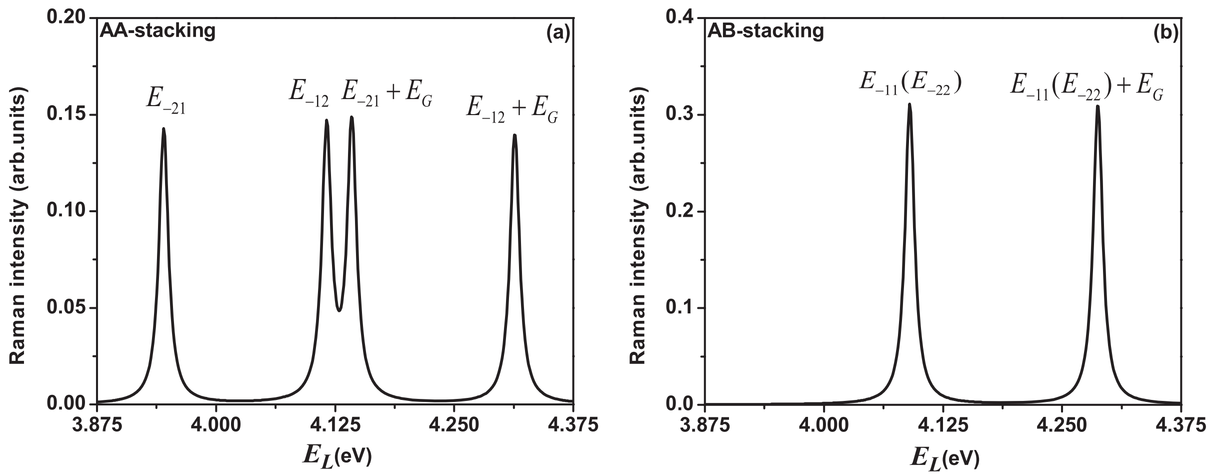

Figure 3. The first-order resonant G-band Raman peak intensity versus the incident laser energy E_L for the undoped (a) AA- stacking BLG and (b) AB-stacking one. Here, E_{-21}, E_{-12}, E_{-11}, E_{-22} denote the optical transition energies. E_G is the G-band phonon energy.

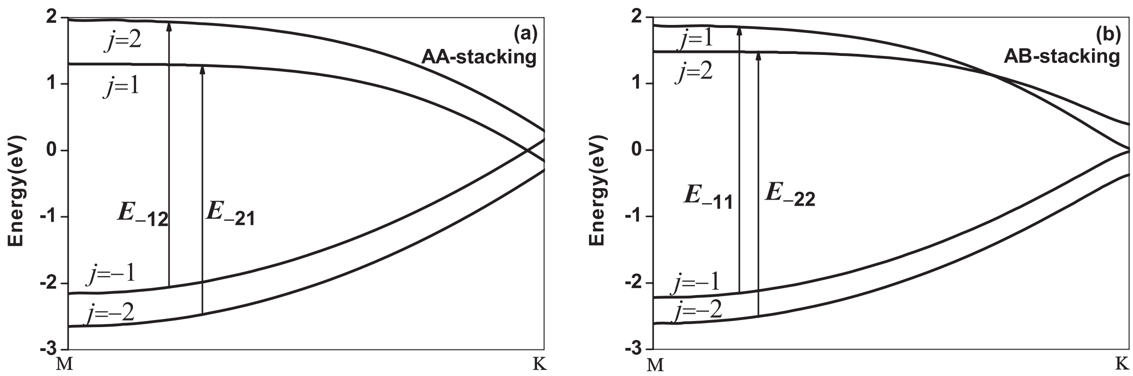

Figure 4. The energy band structures along $M \to K$ for (a) AA-stacking BLG and (b) AB-stacking BLG. Here, the black arrows denote the resonant optical transitions in the first-order resonant Raman spectra.

4. Summary

In summary, we have studied the IR and the first-order G-band resonant Raman spectra of the undoped AA-stacking BLG, using the first-principles method, and made a comparison of them with those of the AB-stacking one. It is found that: (1) the AA-stacking sequence causes a very different band structure from that of the AB-stacking one, so inducing a characteristic jump at 450 meV in its IR spectrum for $\vec{E} \parallel x$, which is different from that at 373 meV for the AB-stacking BLG. On the other hand, when $\vec{E} \parallel z$, a peak exists at 450 meV in the IR spectrum of AA-stacking BLG, which is also different from a jump at 746 meV for the AB-stacking BLG. (2) Its first-order G-band resonant Raman spectra are also found to be greatly different from those of the undoped AB-stacking BLG. For example, the AA-stacking BLG exhibits two pairs of resonant peaks, in contrast to only one pair of the doubly degenerated peaks for the AB-stacking one. Also, the peak intensity of the AA-stacking BLG is found to be nearly half of that for the AB-stacking one.

Therefore, the above-mentioned specific IR and first-order resonant G-band Raman spectra of the undoped AA-stacking BLG, obtained from our calculations, could be used to identify experimentally the AA-stacking BLG from the AB-stacking one, and also from the MLG.

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